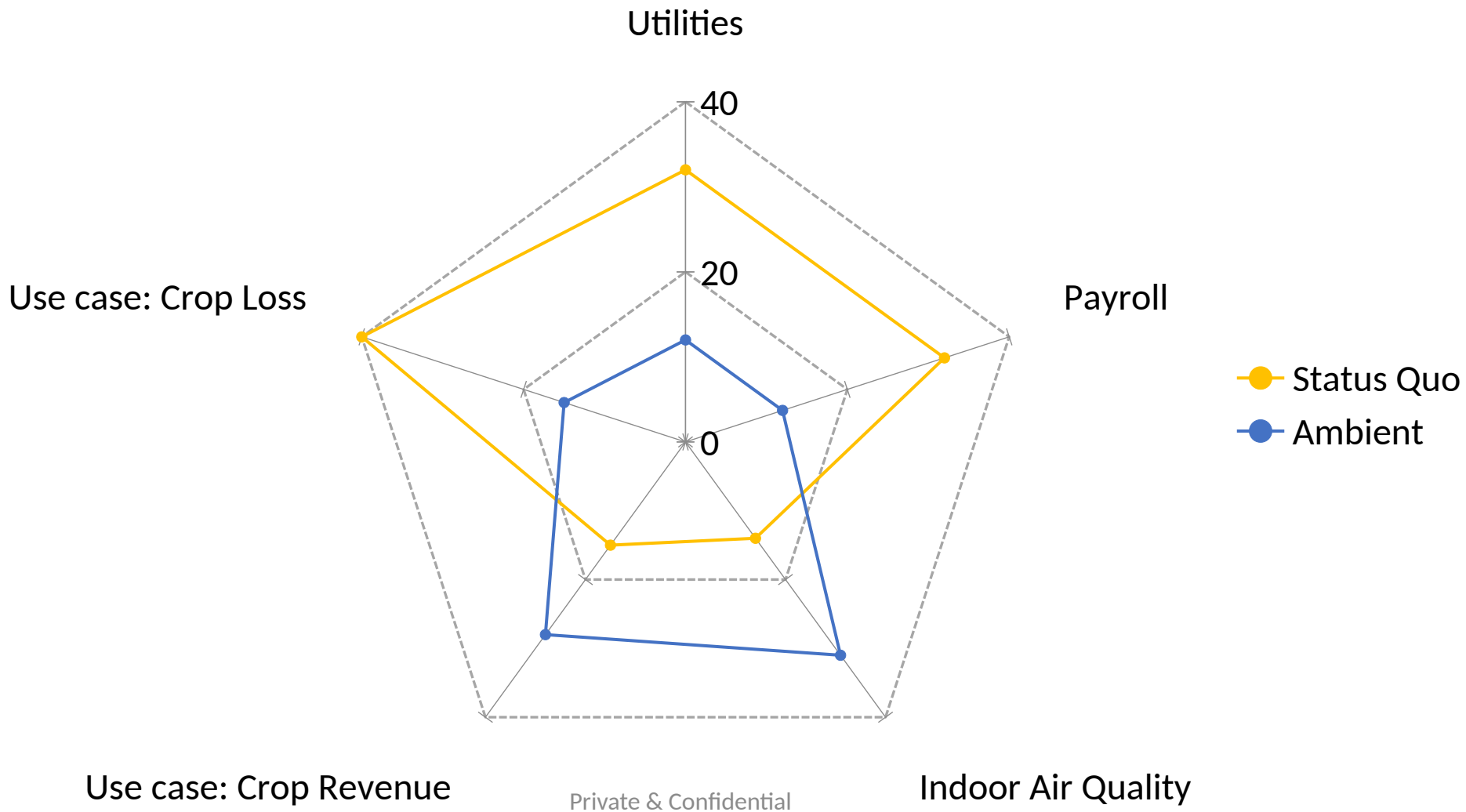
Ambient Systems

Business Plan Technology Slides

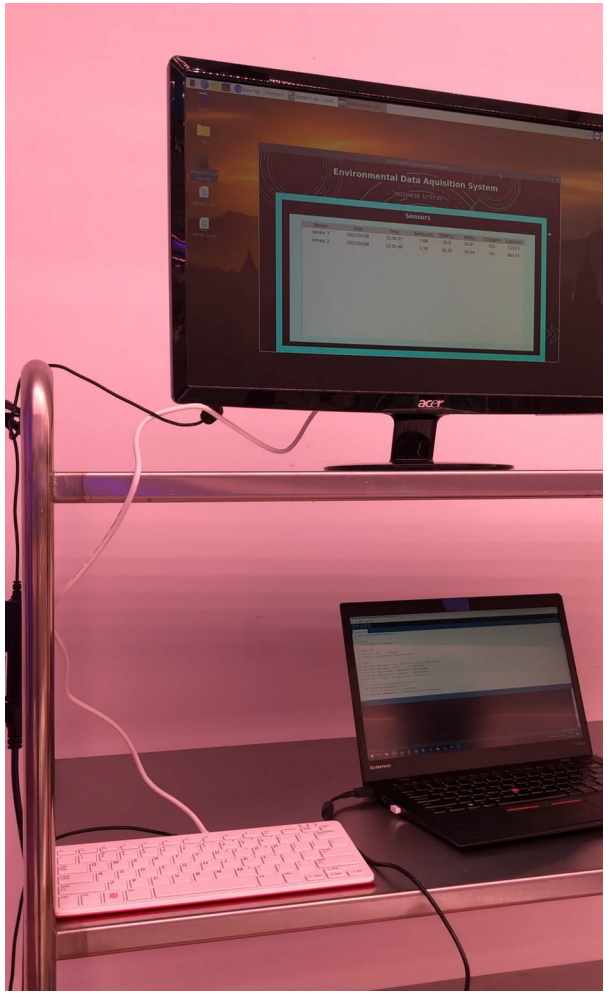
Problem: Multi-objective Optimization



Proof-of-Concept MVP Pilot

1. Collected climate data successfully
2. Identified gap between current and optimal
3. ed s t





Solution Targets Quantified

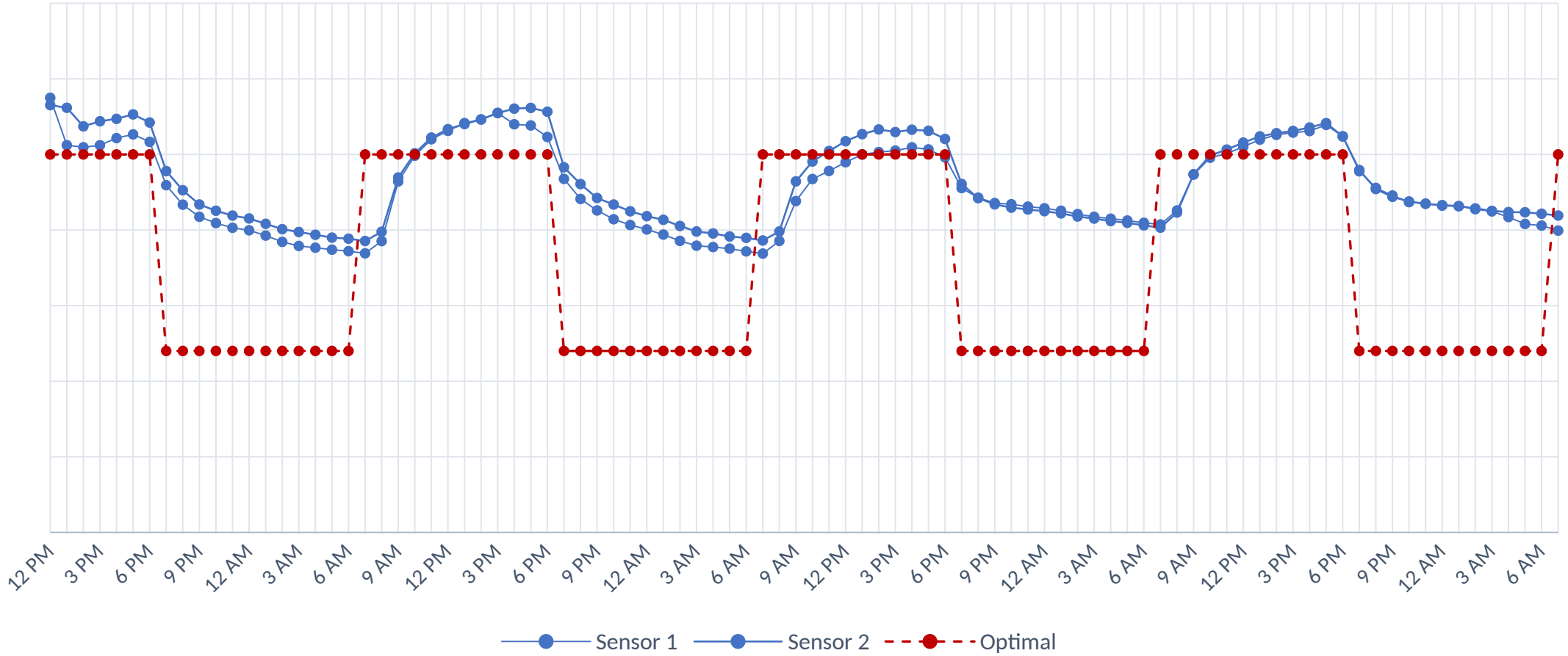
	Parameter	Actual	Optimal	Sample Action
Current POC	Air Temperature	28 C / 20 C	25 C / 12 C	Decrease temperature setpoint Increase ventilation
	Relative Humidity	55% to 70%	60% to 75%	Install fogging machines
	CO2	300 to 700 ppm	500 to 1200 ppm (min >330 ppm)	Decrease/increase ventilation (depends on CO2 concentrations outside) Install CO2 regulator
	Spectrum Light	2300 lux	5000+ lux	Turn off non-farming lights Reduce light intensity

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Full-scale Plot	Soil Temperature	MVP 2.0		
	Soil Moisture			
	pH level			
	Electrical conductivity (EC_w)			
	Crop specific parameters			

Temperature

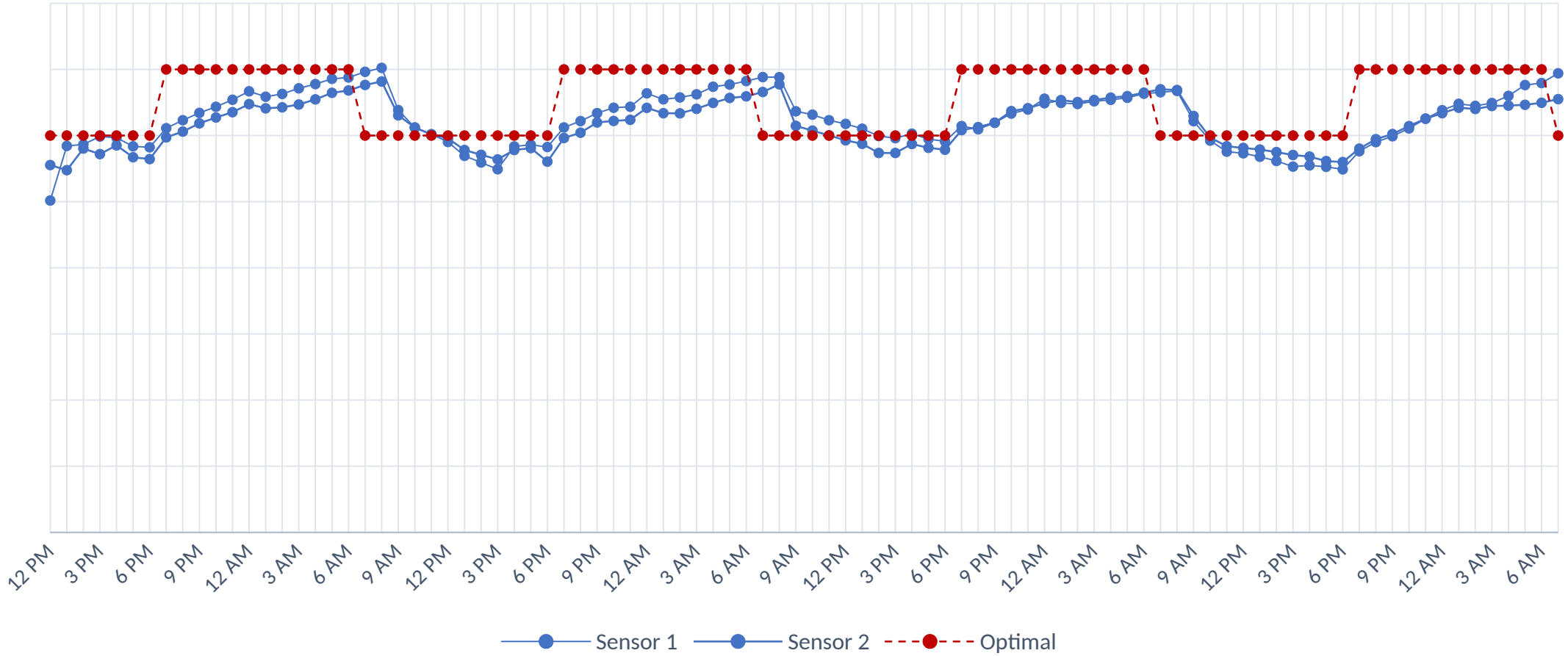
Leaf and petiole growth
Brighter leaf color
Greater sucrose
More biomass at the roots



Temperatures after bloom affect plant growth and fruit quality of strawberry, , 2000

Relative Humidity

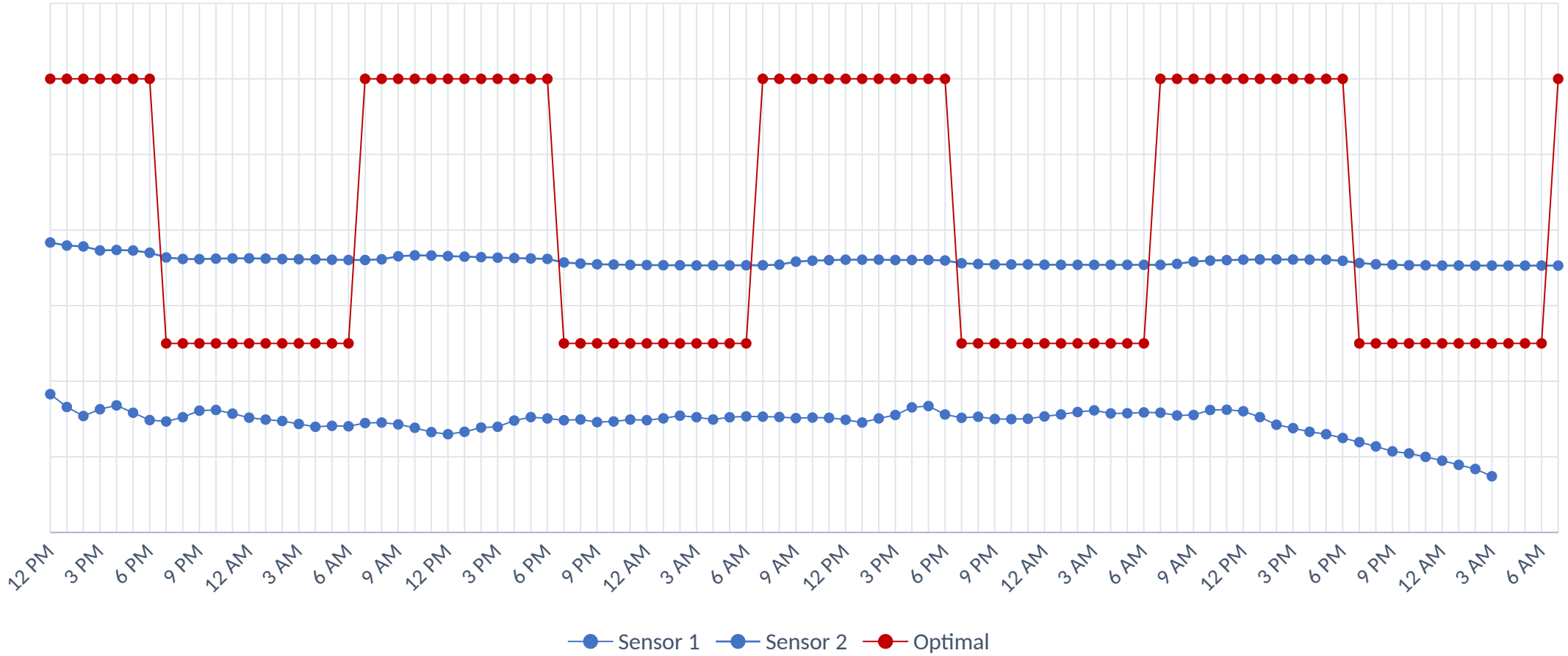
Avoid tip-burn and calyx burn
Avoid fungi
Avoid powdery mildew
Avoid leaf spot (bacterial blight)



Crop Guide: Growing Strawberry, , 2004

CO2

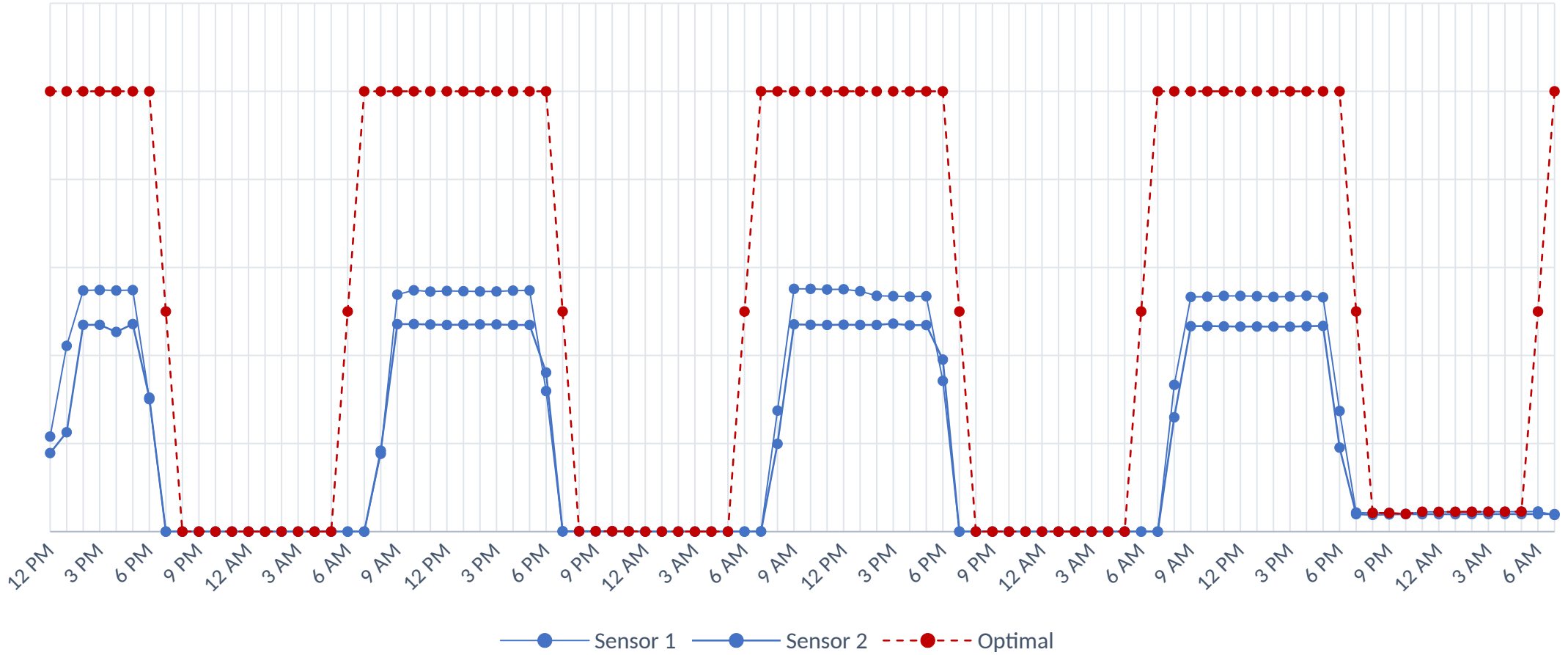
Increased photosynthesis
Earlier flowering
Higher fruit yields
Improved stem strength and flower size



Carbon Dioxide In Greenhouses, Ministry of Agriculture, Food and Rural Affairs, 2021
Carbon dioxide (CO2) applications in indoor growing,

Light Intensity

Increased photosynthesis

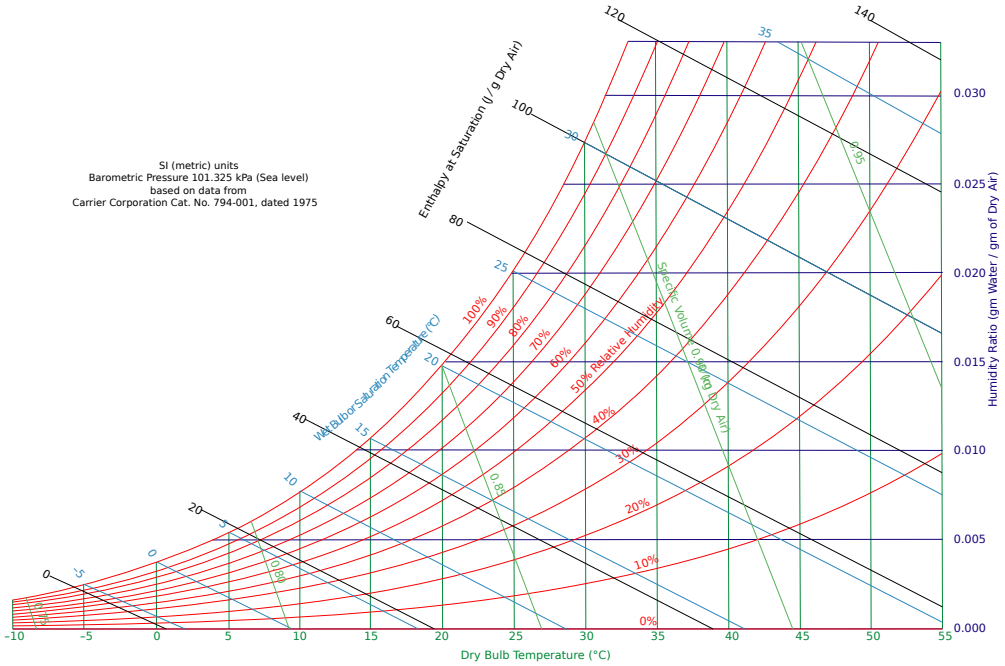


Effect Of Light Intensity, Temperature And Co2 Concentration On Photosynthesis And Growth Of In Vitro, International Society for Horticultural Science, 1989

Why not just change the AC settings?

#1: Interactions

- Psychometrics: temperature and relative humidity
- Light increases temperature and decreases humidity
- Decreasing temperature decreases CO2 concentrations in the air
- Increased ventilation introduces hot humid air f outside that needs to be cooled, purified, e hed, and re-humidified → very energy i sive → very expensive
- Safety: CO2 is needed by the plants, but human workers need oxygen to work indoors
- Photosynthesis process: lights and CO2 interact (both need to be high or both need to be low)
- Infiltration/exfiltration: porous building materials



#2: Equipment

- Building equipment is often outdated
- AC controls are limited: only allows setting temperature and fan speed (not granular enough)
- AC controls cannot be managed remotely
- Need additional or custom equipment based on plant type, growing stage, and outdoor climate



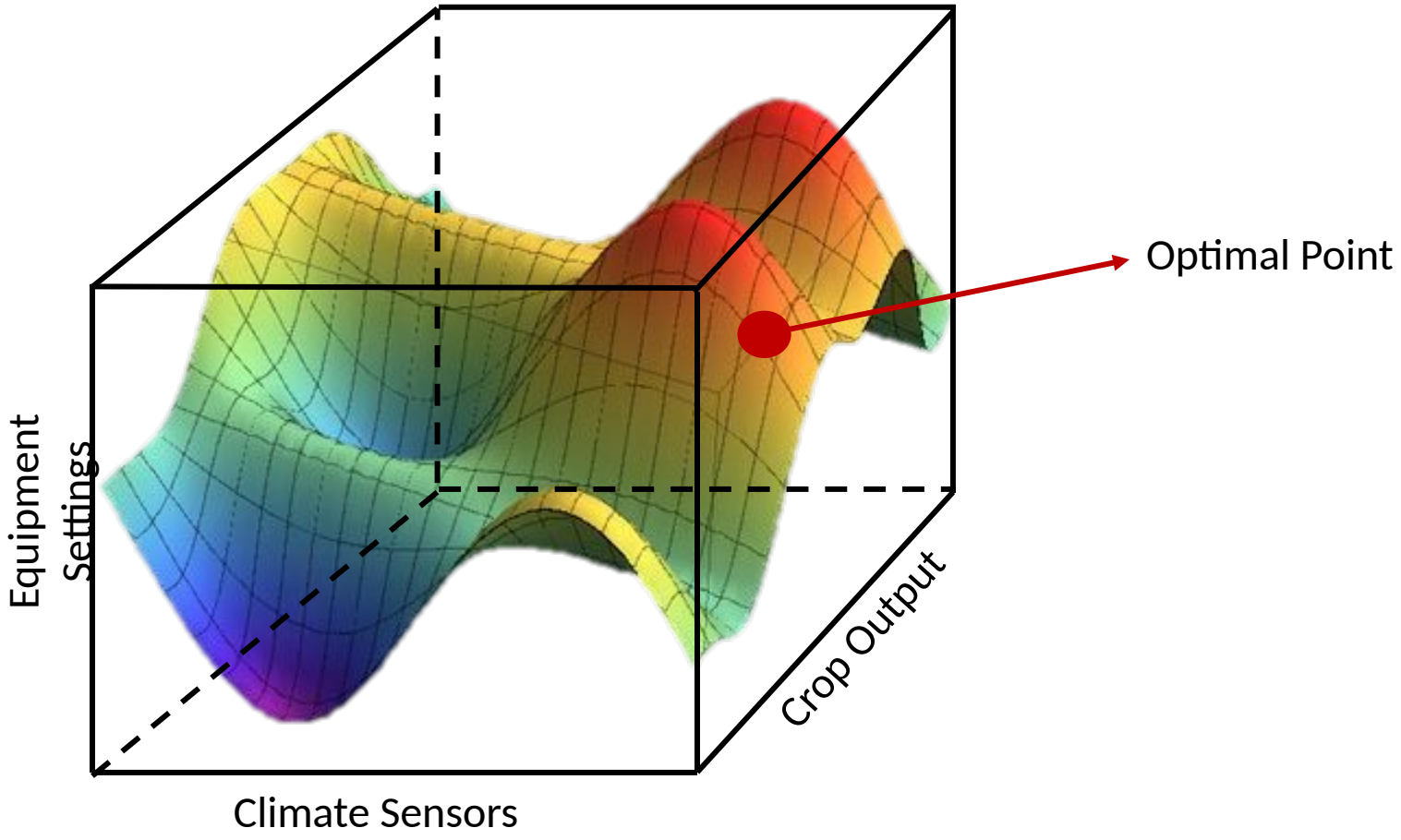




How to address this?

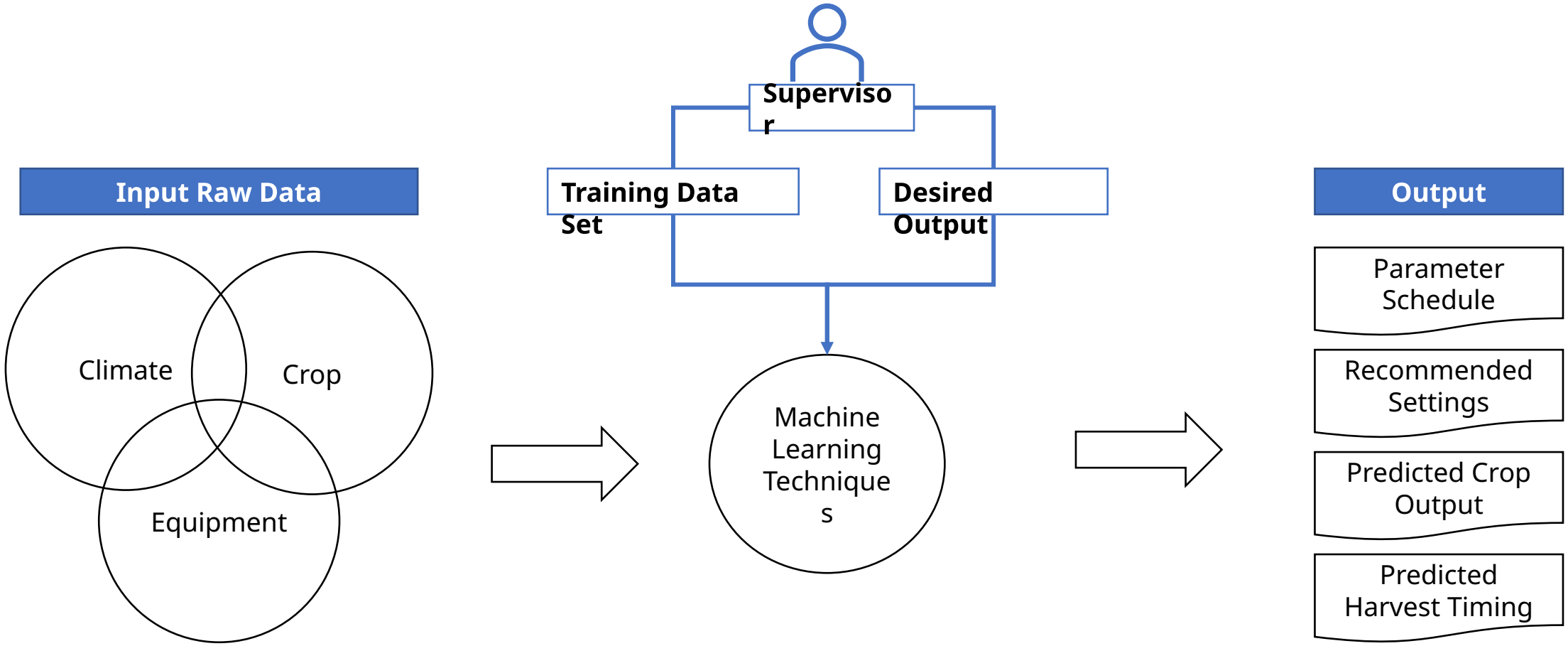
Optimization using building science-based AI models

Optimization: Multi-Dimensional Gradient Descent

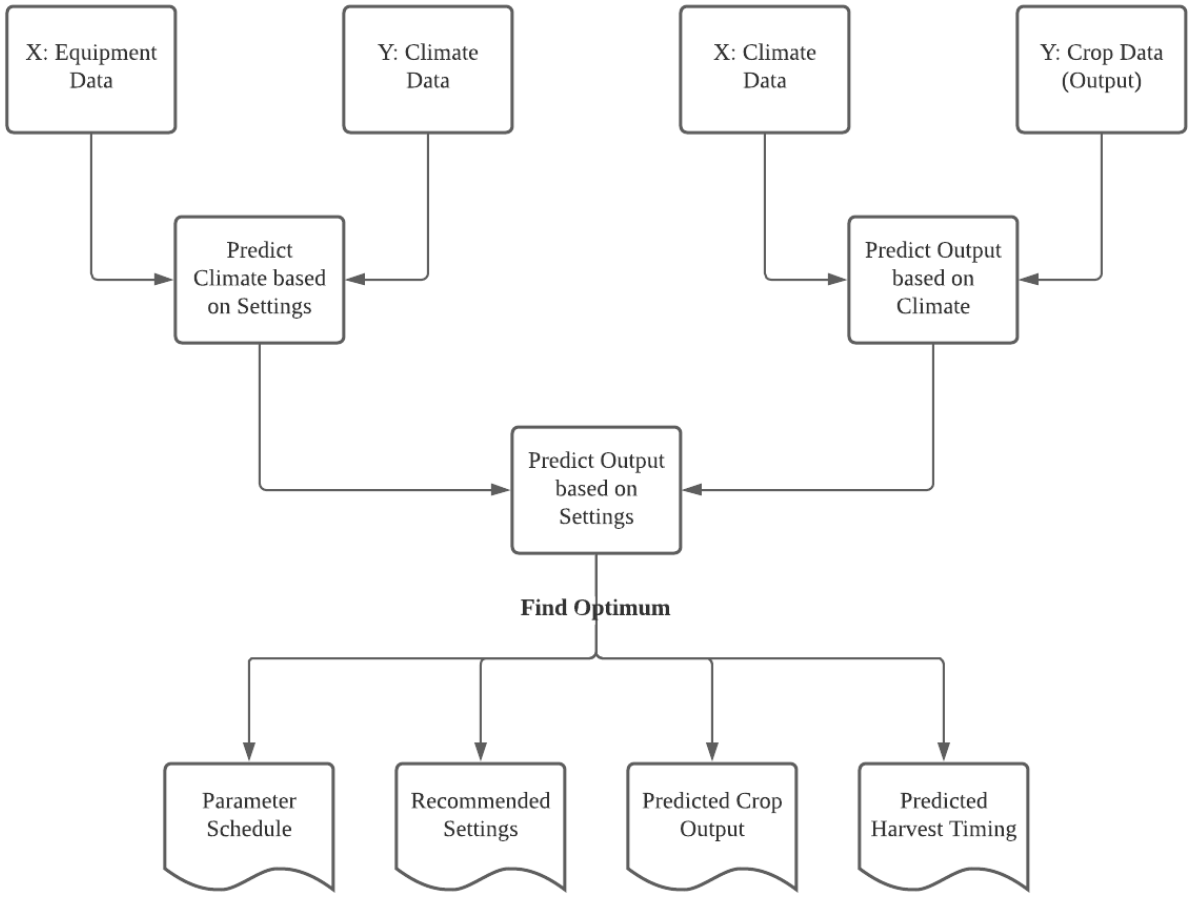


Ambient's Machine Learning Optimization Model

Patent



Supervised Learning: Three Stages



Notes:

- Non-climate parameters are already automated and controlled via dedicated systems (i.e., LED lighting, nutrition, irrigation, etc.)
- How Crop Data (output) is measured will vary based on farm type → it will impact data collected
- Product is agnostic to how crop output is defined or measured (assuming climate is an important input)
- More frequent data collection will increase the training of the feedback loop (e.g., providing crop data daily will allow quicker calibration than providing data at the end of each harvest)
- We can work with the farmer to provide the necessary data acquisition for crop quality (e.g., image recognition) and platform for training (e.g., optimization features)

Patent Classification

A	HUMAN NECESSITIES
B	PERFORMING OPERATIONS; TRANSPORTING
C	CHEMISTRY; METALLURGY
D	TEXTILES; PAPER
E	FIXED CONSTRUCTIONS
F	MECHANICAL ENGINEERING; LIGHTING; HEATING; WEAPONS; BLASTING
G	**PHYSICS** **G06:** COMPUTING; CALCULATING OR COUNTING **G06N:** COMPUTER SYSTEMS BASED ON SPECIFIC COMPUTATIONAL MODELS **G06N 3/00**: Computer systems based on biological models **G06N 5/00**: Computer systems using knowledge-based models **G06N 7/00**: Computer systems based on specific mathematical models **G06N 20/00**: Machine learning
H	ELECTRICITY

What are the benefits?

Productivity & Operational Efficiency

Ambient's optimization models show you how to create an optimal environment with:

- No need to manage sensors or raw data
- No need to manually adjust ACMV equipment early in the morning and late in the afternoon to accommodate plant growing schedule
- No need to manually adjust built-in (non-farming) lights
- No need for analysis of complex interactions between parameters
- No need to wait until seeing harvest results to fine-tune the parameters
- No need to deal with complex thermal inertia calculations
- No need for complex set-ups to perform farming experiments
- No need to spend time evaluating which mechanical equipment to buy (we recommend it)
- No need to worry about finding and selecting inter-operable systems (we shortlist them for you)

How much time can Ambient save you?

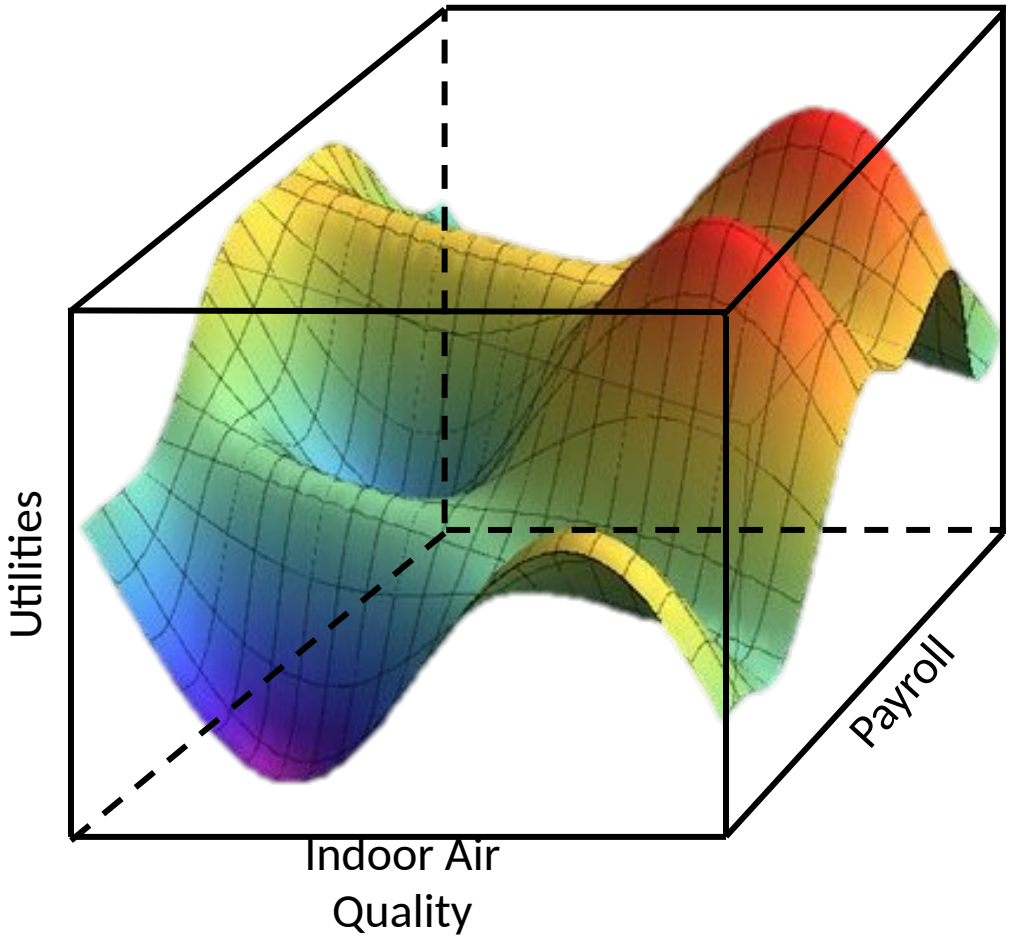
Pilot Outcomes

Recommendations

Insight	**Recommendation**	**Cost**
Variability between days	Monitor for longer period of time (entire harvest cycle)	Ambient Platform Tier 1
No utilities data	Monitor energy consumption	$5k-$10k hardware modification Ambient Platform Tier 1
High temperature	Decrease daytime and night-time temperature	$5k - $10k portable AC
Low CO2	CO2 regulator (increase CO2)	$1k - $2k CO2 regulator $500 monthly CO2 supply
Moisture timing	Introduce mist earlier in the day Reduce mist later in the day	Ambient Platform Tier 1
Plant with various optimal parameter schedules	Determine farm portfolio optimal (not just for one plant but entire batch)	Ambient Platform Tier 1

Appendix

Optimization: Multi-Dimensional Gradient Descent



Optimization: Weighted Loss Function

Function =

w_crop_value * delta_crop

w_electricity * delta_electricity

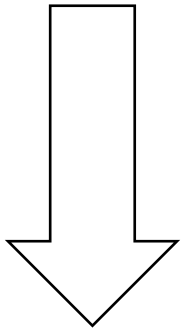
w_labour * delta_labour

Data

- Environmental Data
- Equipment Data
  - ACMV System
  - Lighting System
  - Auxiliary Devices (e.g., micro-fans)
- Use-case Specific Data (e.g., Indoor Farming Data)
  - Crop images
  - Water + Nutrients: pH level, nitrogen, phosphorous, potassium, calcium, sulphur, and magnesium
  - Substrate

Climatic Data

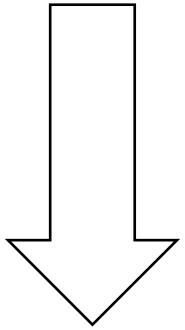
Timestamp	Temperature	Relative Humidity



Timestamp	Label	Temperature	Relative Humidity	CO2	Light Intensity

ACMV System Data

Timestamp	Temperature Setpoint	Ventilation Setpoint

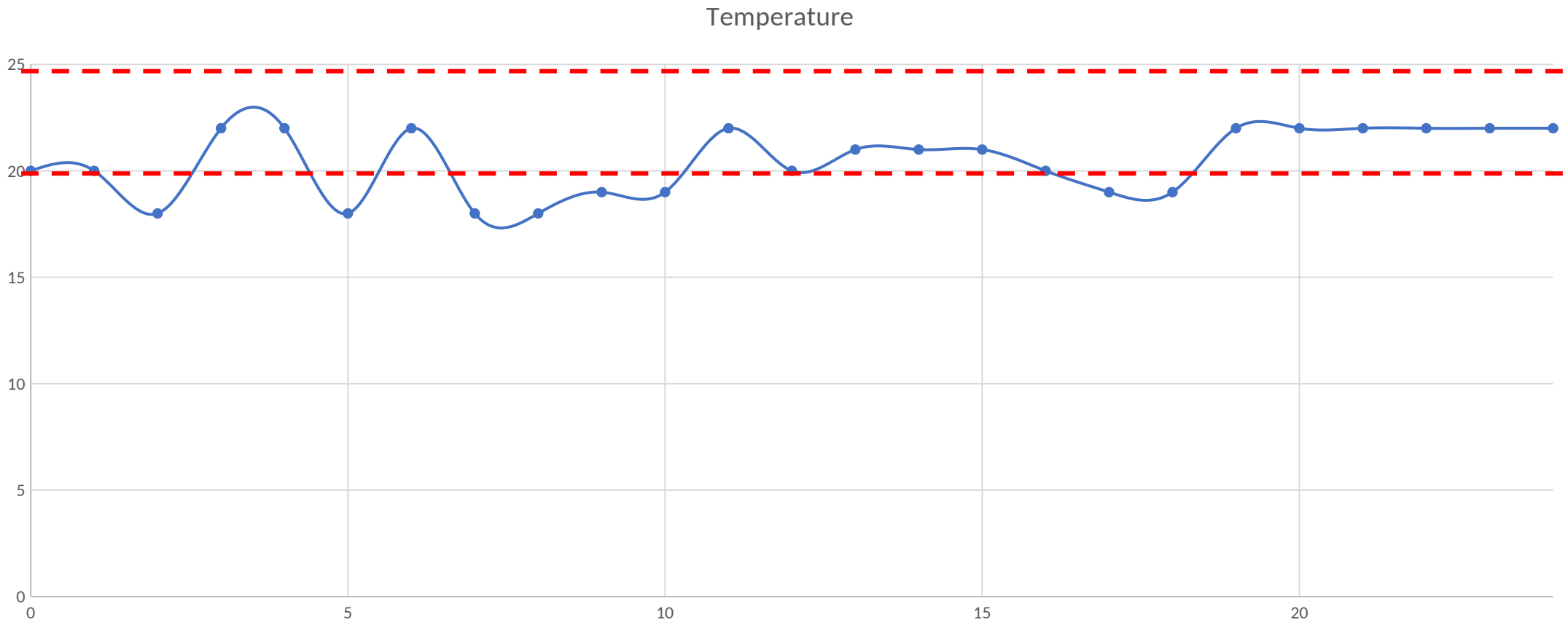


Timestamp	AHU: Supply Temp	AHU: Output Temp	Duct Airflow	Ventilation

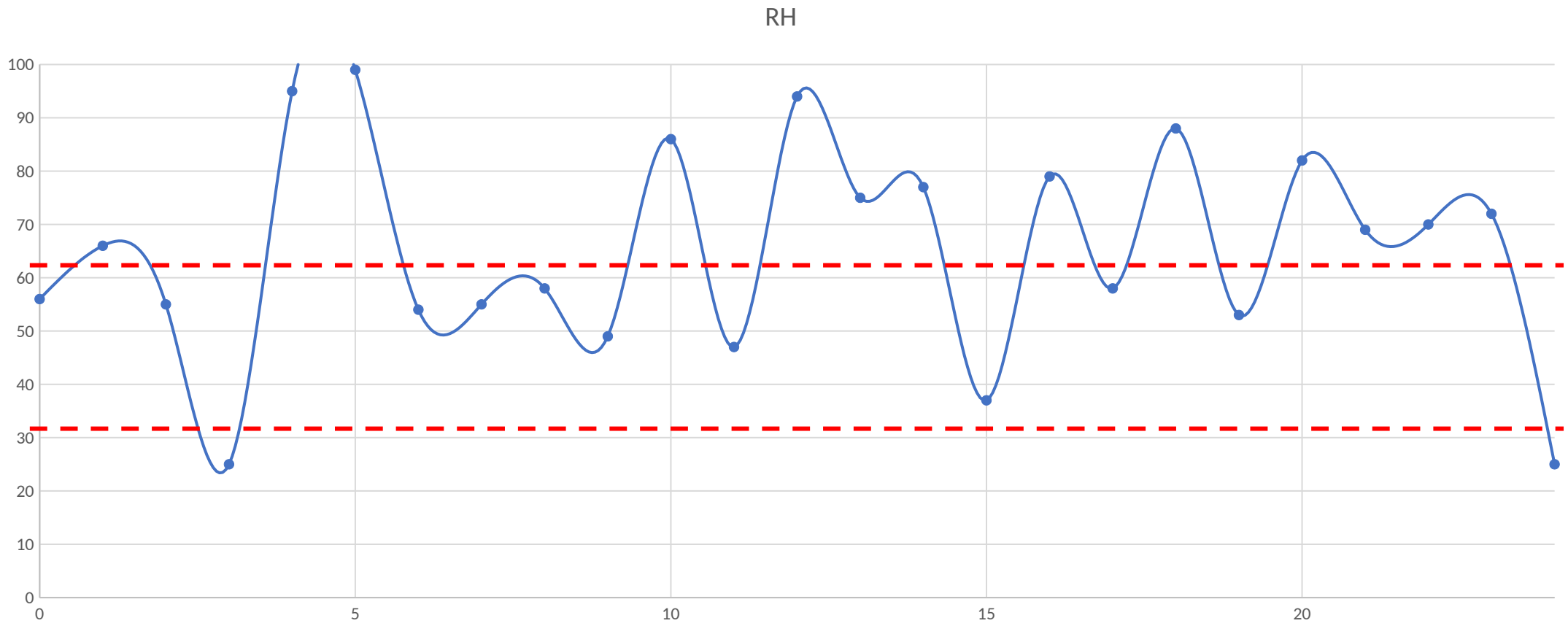
Technology Strategy

- Farmers are either not collecting or not fully utilizing their climate data
- Incentivize adoption through ML features

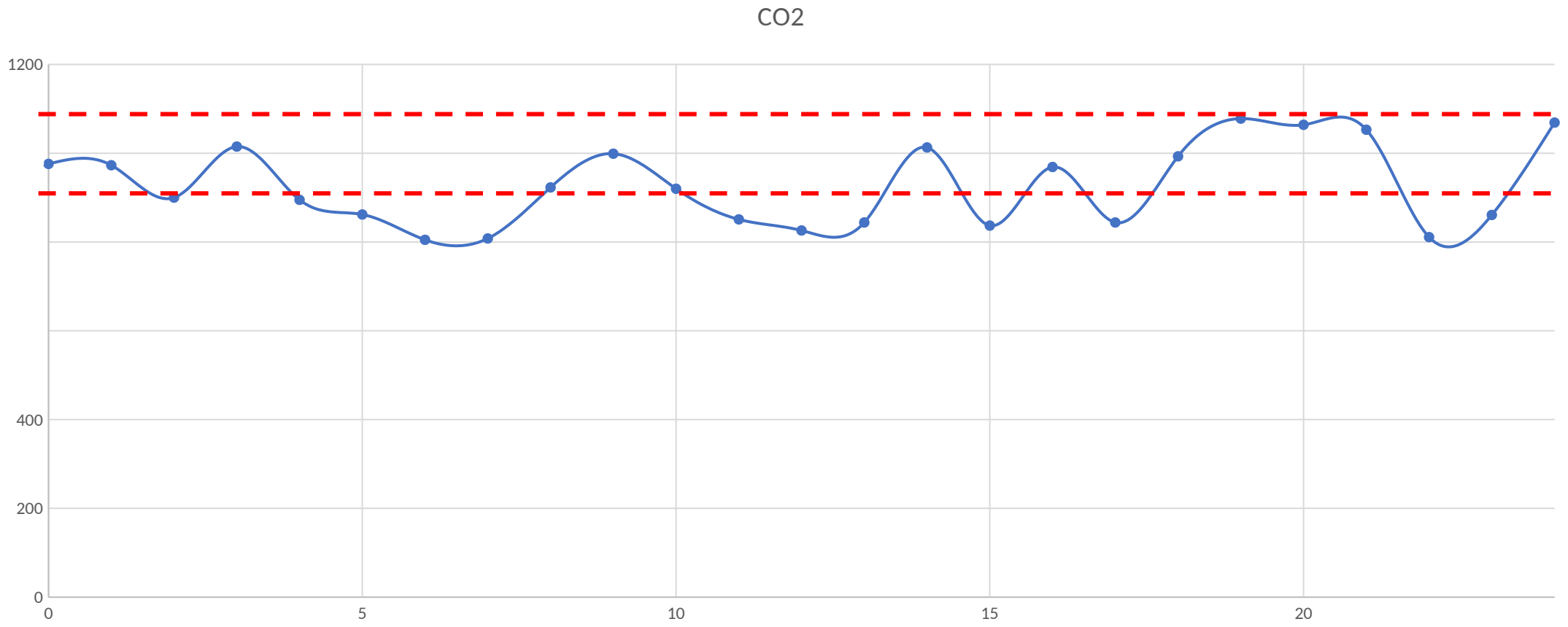
Parameter: Temperature



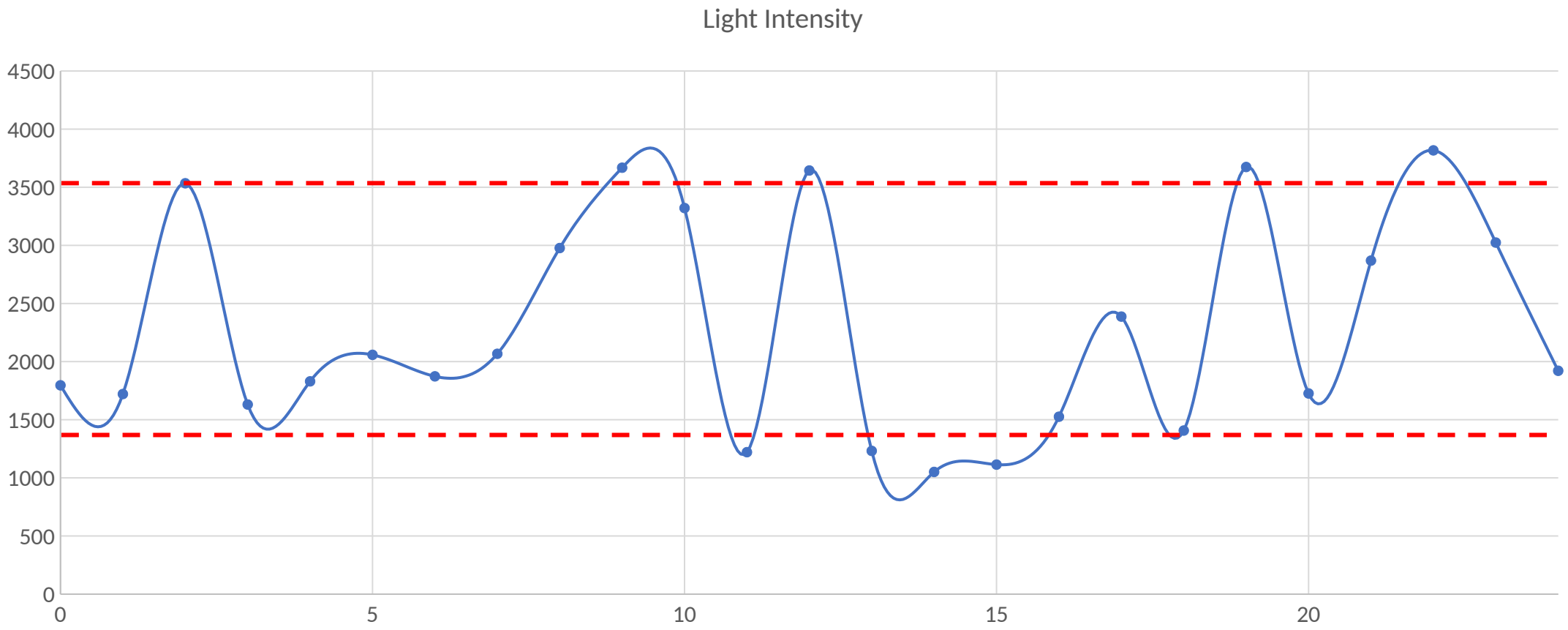
Parameter: Relative Humidity



Parameter: CO2



Parameter: Light



Investment

- $50k gets us to:
  - Completing pilot
  - Completing IP
  - Pre-seed round ($100k)
  - Headcount (~3-4): founders, engineering, sales and partnerships
  - October 2021
- $100k ($50k + $50k):
  - Headcount (~5): founders, engineering, project management, sales and partnerships
  - March 2022
- $2M gets us to:
  - $300k ARR (~10 container sized farming customers)
  - Headcount (~15)

Q&A: Technology

Q: What is the IP?

A: IP is the optimization method that uses machine learning. The plan is to classify the patent under WIPO class G (PHYSICS) and more specifically G06N (COMPUTER SYSTEMS BASED ON SPECIFIC COMPUTATIONAL MODELS)

Q: What specifically are you collecting and what are you optimizing?

A:

Q: How much data do you have?

A: We have three weeks of environmental data from Singrow so far. Additional data collection requires additional hardware, which can be expensive. We are planning to cover these costs using a combination of product fees and GRIP funding.

Q: What have you discovered so far?

A:

Q: What if the model doesn't work?

A: We intuitively know that climate impacts the outputs. However, it is possible to find out that there is no relationship between parameters and crop output (i.e., prediction is no better than chance). Moreover, there are other important outputs that we will be optimizing for.

Q: What if someone copies it?

A: We understand that the IP at this stage can be improved. Through working with customers, collecting additional data, and testing the models, we believe we can greatly enhance it. Moreover, we see an opportunity to bring in necessary expertise across NUS to refine certain aspects of the technology. For instance, we are looking to bring in AgTech experts to help us understand the relationship among climate and crop output variables. Similarly, e machine learning experts to advise us on and selection.

Q&A: Business Model

- Q: What will it cost them to implement the full solution. Only revenue is subscription?
- A:

Q&A: Funding

- Q: Where does $X get you?
- A:
- Q: How much market penetration do you need to get these revenues?
- A: